

MACHINE LEARNING WITH PYTHON

Examination Solutions (E1PY210B)

Galgotias University | SCAT | April 2026

Section A: Python Fundamentals

1. What are keywords and identifiers in Python? [cite: 240]

Keywords: Reserved words that have special internal meanings to the Python interpreter (e.g., if, while, return). They cannot be used as variable names. [cite: 240]

Identifiers: Names used to identify variables, functions, classes, or modules. They must start with a letter or underscore and are case-sensitive. [cite: 240]

4. Python program for List Operations [cite: 240]

```
# List of 10 integers
numbers = [12, 45, 7, 23, 56, 89, 34, 1, 90, 10]

# Calculations
max_val = max(numbers) # Returns the largest item
min_val = min(numbers) # Returns the smallest item
avg_val = sum(numbers) / len(numbers) # sum() adds all items;
len() counts them

print(f"Max: {max_val}, Min: {min_val}, Average: {avg_val}")
```

Section B: AI & Model Evaluation

2. Artificial Intelligence vs. Machine Learning [cite: 240]

AI: The broad concept of machines being able to carry out tasks in a way that we would consider "smart" (e.g., Chess-playing computers). [cite: 240]

ML: An application of AI based around the idea that we should just give machines access to data and let them learn for themselves (e.g., Email spam filters). [cite: 240]

6. Confusion Matrix Calculations [cite: 240]

Given: TP=50, FP=10, FN=20, TN=120 [cite: 240]

- **Accuracy:** $(TP+TN)/(Total) = (50+120)/200 = \mathbf{0.85 (85\%)}$ [cite: 240]
- **Precision:** $TP/(TP+FP) = 50/(50+10) = \mathbf{0.833 (83.3\%)}$ [cite: 240]
- **Recall:** $TP/(TP+FN) = 50/(50+20) = \mathbf{0.714 (71.4\%)}$ [cite: 240]

Section C: Data Preprocessing

7. Dataset Cleaning and Preparation [cite: 244, 245]

Following the instructions for the provided student dataset: [cite: 245]

1. **Unnecessary Column:** Student_ID should be removed as it has no predictive power. [cite: 246]
2. **Encoding:** Convert Gender (Male: 0, Female: 1) and Final_Result (Pass: 1, Fail: 0). [cite: 249]
3. **Scaling:** Apply Min-Max Scaling to Attendance and Previous_Marks. [cite: 250]

Cleaned Sample Entry: [Gender: 0, Attendance: 0.85, Study_Hours: 3, Prev_Marks: 0.78, Result: 1] [cite: 244, 255]